

PBL Evaluation Scheme

- Problem identification – 15%
- Literature survey & analysis – 15%
- Solution design & innovation – 20%
- Prototype/implementation – 20%
- Presentation & communication – 10%
- Teamwork & progress – 10%
- Reflection & documentation – 10%

PBL structure

UNIT I – Problem Identification & Planning

- Introduction to PBL
- Group Formation
- Problem Identification
- Topic Finalization
- Literature Survey
- Existing System Analysis
- Requirement Gathering
- Feasibility Study
- Technology Selection
- Synopsis Submission

UNIT II – Solution Development

- Solution Design
- Prototype Development
- Development Phase
- Testing
- User Feedback
- Improvement
- Intermediate Presentation

UNIT III – Reflection & Documentation

- Final Implementation
 - Project Demonstration
 - Reflection on Learning
 - Challenges & Solutions
 - Future Scope
 - Report Writing
 - Final Presentation
 - Viva
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PBL Report Index – BCA

Preliminary Pages

1. **Cover Page**
 2. **Certificate**
 3. **Declaration**
 4. **Acknowledgement**
 5. **Abstract / Executive Summary**
 6. **Table of Contents**
 7. **List of Figures**
 8. **List of Tables**
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Chapter 1 – Introduction and Problem Identification

1.1 Introduction

Brief introduction to the project and the domain.

1.2 Problem Identification

Explain the real-world problem identified by the students.

1.3 Problem Statement

Clearly state the specific problem the project aims to solve.

1.4 Motivation

Why did the students select this problem?

1.5 Objectives

List the objectives of the proposed solution.

1.6 Scope of the Project

What is included and what is outside the scope?

1.7 Target Users / Stakeholders

Who will use or benefit from the proposed solution?

1.8 Expected Learning Outcomes

What did students expect to learn from the project?

Chapter 2 – Problem Investigation and Existing System

2.1 Literature Survey

Study of existing research, applications, articles, etc.

2.2 Existing Systems / Solutions

Discuss currently available solutions.

2.3 Limitations of Existing Solutions

Identify gaps or problems in existing solutions.

2.4 User / Stakeholder Study

Survey, interviews, questionnaires, observations, etc.

2.5 Analysis of Findings

What did students learn from the investigation?

2.6 Proposed Solution

Explain the proposed solution.

Chapter 3 – Requirements and Feasibility Analysis

3.1 Functional Requirements

What should the system do?

3.2 Non-Functional Requirements

Performance, usability, security, reliability, etc.

3.3 Hardware Requirements

3.4 Software Requirements

3.5 Technology Stack

Programming language, database, framework, tools, etc.

3.6 Feasibility Study

- Technical feasibility
 - Economic feasibility
 - Operational feasibility
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Chapter 4 – Solution Design

4.1 Proposed System Architecture

4.2 System Workflow

4.3 Use Case Diagram

4.4 Data Flow Diagram / Activity Diagram

4.5 Database Design

- ER Diagram
- Database tables

4.6 User Interface Design

Wireframes/mockups/screens.

4.7 Module Description

Chapter 5 – Prototype / Implementation

This is where students demonstrate that they **actually implemented their proposed solution**.

5.1 Prototype Development

5.2 Development Environment

5.3 Module-wise Implementation

For example:

5.3.1 Login Module

5.3.2 User Module

5.3.3 Admin Module

5.3.4 Report Module

5.4 Core Functionality

Explain the **minimum essential features** developed to demonstrate the solution.

5.5 Screenshots of Implementation

5.6 Challenges Faced During Development

5.7 Solutions / Improvements Made

Chapter 6 – Testing and Evaluation

6.1 Testing Strategy

6.2 Test Cases

Test Case	Input	Expected Output	Actual Output	Status
Login	Valid credentials	Login successful	Login successful	Pass
Login	Invalid credentials	Error message	Error message	Pass

6.3 Functional Testing

6.4 User Feedback

If students showed the prototype to users/faculty, record the feedback.

6.5 Evaluation of Proposed Solution

Did the solution actually address the identified problem?

6.6 Improvements Based on Feedback

This section is **particularly important for PBL**.

Chapter 7 – Results and Learning

7.1 Results

What was achieved?

7.2 Problem-Solution Evaluation

Explain how the developed solution addresses the original problem.

7.3 Learning Outcomes Achieved

What technical and non-technical skills did students gain?

7.4 Challenges and Lessons Learned

7.5 Teamwork and Individual Contributions

7.6 Reflection on Learning

Students should explain:

What did we learn from this project that we did not know before?

This is one of the sections that helps distinguish PBL from a conventional Mini Project report.

Chapter 8 – Conclusion and Future Scope

8.1 Conclusion

8.2 Limitations

8.3 Future Enhancements

8.4 Possibility of Real-World Deployment

References

Books, research papers, websites, documentation, APIs, etc.

Appendices

Appendix A – Questionnaire

If a survey was conducted.

Appendix B – Additional Screenshots

Appendix C – Sample Code

Only important code snippets, **not necessarily the entire source code**.

Appendix D – User Feedback

Appendix E – Individual Contribution

Recommended Final Index

For your **MIT-WPU BCA PBL**, I would actually simplify it to the following **8-chapter structure**:

Chapter Title

1 Introduction & Problem Identification

Chapter Title

- 2** Problem Investigation & Existing System
- 3** Requirements & Feasibility Analysis
- 4** Proposed Solution & System Design
- 5** Prototype / Implementation
- 6** Testing, Feedback & Evaluation
- 7** Results, Learning & Reflection
- 8** Conclusion & Future Scope

References

Appendices

Why I recommend this structure

It directly follows the PBL journey:

**Problem → Investigation → Requirements → Solution → Prototype → Testing → Feedback
→ Learning → Conclusion**